

ROOSTING ECOLOGY OF THE NORTHERN BAT (*MYOTIS SEPTENTRIONALIS*) AND COMPARISONS WITH THE ENDANGERED INDIANA BAT (*MYOTIS SODALIS*)

RODNEY W. FOSTER AND ALLEN KURTA

Department of Biology, Eastern Michigan University, Ypsilanti, MI 48197

During 1993 and 1994, we radiotracked 11 adult female and juvenile northern bats (*Myotis septentrionalis*) to 32 roost trees: 18 silver maples (*Acer saccharinum*), 1 red maple (*A. rubrum*), and 13 green ashes (*Fraxinus pennsylvanica*). Fifty-three percent of the trees were living; 52% of the roosts were in crevices or hollows, and the rest were under exfoliating bark. Northern bats did not use American elms (*Ulmus americana*) in proportion to their abundance, but in general, characteristics of roost trees were similar to a random sample of apparently suitable trees. Northern bats changed roosts every 2 days, and distance between roosts varied from 6 to 2,000 m. As many as 60 adults were found in a single tree, making this the largest summer aggregation ever reported for the species. Compared with Indiana bats (*M. sodalis*) separately studied at the same site, northern bats moved greater distances between roosts and roosted more often in maples, cavities, living trees, and areas with high canopy cover.

Key words: *Myotis septentrionalis*, northern bat, *Myotis sodalis*, Indiana bat, behavior, roosts, endangered species, Michigan

Appropriate shelter is a necessity for small terrestrial vertebrates, providing them with refuge from predators and protection from extremes in temperature or precipitation. For small mammals and birds, appropriate shelter also is needed for raising dependent young. Recent development of miniature (<1 g) radiotransmitters has allowed biologists to determine that many, perhaps the majority, of small (5–20 g) insectivorous bats of temperate North America roost in trees during summer, often in hollows or crevices (Barclay and Brigham, 1996; Vonhof and Barclay, 1996). The habit of roosting in tree hollows, similar to those occupied by birds and squirrels (Baker, 1983; Brewer et al., 1991), makes one wonder how sympatric species of bats partition a resource that already is in demand by birds and rodents. In addition, many bats roost underneath exfoliating bark of dead trees (Barclay and Brigham, 1996; Kurta et al., 1993a, 1996). The prevalence of this habit among different species of bats, and

at least one species of bird (Kurta and Foster, 1995) also suggests that different organisms may partition a potentially limiting resource (Humphrey, 1975; Tidemann and Flavel, 1987). Anecdotal evidence indicates that birds do occasionally compete, directly or indirectly, with temperate bats for roosts (Maeda, 1974; Mason et al., 1972; van Heerdt and Sluiter, 1965), and some tree-roosting bats often move into manmade bird houses or bat houses, which suggests that suitable trees may not be sufficiently available in all areas at all times (Gerrell-Lundberg and Gerrell, 1994; Harmata, 1969; Maeda, 1974; Tuttle and Hensley, 1993).

Despite the possible niche overlap and potential for limited roosts, there have been no studies comparing the roosting or nesting requirements of North American bats and other taxa, and there have been few attempts to compare the requirements of different species of tree-roosting bats. The difficulty and expense of capturing small bats

and radiotracking them has led most researchers to either concentrate on a single species (Brigham et al. 1997; Callahan et al., 1997; Kalcounis and Hecker, 1996; Lunney et al., 1988; Ormsbee, 1996) or radiotrack whatever bats are captured (Taylor and Savva, 1988; Vonnhof and Barclay, 1996). The first approach makes interspecific comparisons difficult because of varying methodology, habitat, and geography, and the second approach limits interspecific comparisons because of small samples. Vonnhof and Barclay (1996), for example, compared roosts of five groups of bats in western Canada, but average number of roosts per taxon in their study was only four.

The northern bat, *Myotis septentrionalis*, is a small (6–9 g) insectivorous bat that lives in eastern North America, ranging from southern Canada to Florida (Barbour and Davis, 1969). Northern bats hibernate in abandoned mines or caves in winter and then disperse throughout their range in spring. Females are colonial in spring and summer and gather in small groups to give birth and raise their single young to maturity, before returning to their hibernacula in September for mating. Prior to our study, little was known concerning the roosting habits of females and young during the maternity period. Virtually all information on the roosting ecology of the northern bat in summer came from serendipitous finds of single roosts, such as a park shelter (Brandon, 1961) and a barn (Cope and Humphrey, 1972). Observations of northern bats in manmade structures, however, were rare, which indicated that these bats did not use buildings consistently as roosts and that trees were the most likely location of maternity roosts (Mumford and Cope, 1964; Sasse and Pekins, 1996).

Our objective was to describe the roosting niche of the northern bat during the maternity period and various aspects of roosting behavior, such as the frequency and distance of movements. We also compared the ecology and behavior of the northern bat

with those of the federally endangered Indiana bat (*M. sodalis*), another small-bodied (7–10 g), insectivorous, tree-roosting species that occurs in sympatry with the northern bat throughout much of the eastern United States (Barbour and Davis, 1969). Kurta et al. (1993a, 1996) documented the roosting ecology of the Indiana bat in summer in southern Michigan. The present study of the northern bat, performed in the same area with the same techniques and overlapping in time, allowed us to quantify similarities and differences in roosting requirements of these two bats. Such a comparison enabled us to test the hypothesis of Vonnhof and Barclay (1996:1802) that characteristics of roost trees are “similar regardless of the species” of bat.

MATERIALS AND METHODS

Field work was conducted ca. 5 km E of Vermontville, Eaton Co., Michigan, from late May to August 1993 and 1994. Land surrounding Vermontville originally consisted of medium-textured ground moraine, with upland forests of beech-sugar maple (*Fagus grandifolia*–*Acer saccharum*) and swamp forests dominated by trees such as American elm (*Ulmus americana*), green ash (*Fraxinus pennsylvanica*), silver maple (*A. saccharinum*), and swamp white oak (*Quercus bicolor*—Albert et al., 1986). Most of the area currently is a highly fragmented agricultural landscape, dotted with islands of deciduous woodlands and an extensive, albeit narrow, strip of forest along the Thornapple River and its tributaries.

We primarily attempted to capture northern bats in mist-nets while they foraged over the Thornapple River, but on one night in each year, we also netted a flyway between a heavily used roost tree and the river. In either case, bats were removed immediately from the net, and sex, age (juvenile versus adult), and reproductive condition were recorded. Age was assessed by degree of epiphyseal ossification. Pregnancy was determined by palpation, and lactation versus postlactation was determined by condition of teats and ability to express milk.

We attached a miniature radiotransmitter (BD-2B, Holohil Systems Ltd., Carp River, Ontario, Canada) to 12 of 27 northern bats that we cap-

tured (Foster, 1997). After removing a patch of fur from the mid-scapular region with scissors, we glued the transmitter to the skin using a surgical adhesive (Skin-Bond Cement, Smith and Nephew United, Inc., Largo, FL). Each radiotagged bat was punch-marked (Bonaccorso and Smythe, 1972) for future recognition, although northern bats that were not tracked were either punch-marked or banded (Lambournes Ltd., Middlesex, England). Each radiotagged bat was released near the site of capture and radiotracked to its roost the following morning, using a receiver (TRX-2000S, Wildlife Materials, Carbon-dale, IL) and a three- or five-element Yagi antenna. After locating a roost, we estimated population size by counting bats as they emerged during the first 50 min after sunset.

Each tree was designated by a letter representing the order in which it was discovered within a particular year, followed by a two-digit number indicating the original year of discovery; for example, roost B93 was the second roost discovered in 1993. To characterize the habitat, we established 0.1-ha circular plots centered on each roost and identified all trees ≥ 10 cm in diameter at breast height. We ranked each tree according to the amount of loose and peeling bark that remained, based on a system developed for the Indiana bat (Gardner et al., 1991). A tree with $\geq 25\%$ of loose and peeling bark was ranked high; $\geq 10\%$ to $< 25\%$, medium; and $> 0\%$ to $< 10\%$, low. A rating of zero denoted total unsuitability, either because the tree lacked bark or because bark had not yet loosened from the trunk. We also documented presence of cavities and if the tree was dead or alive. If $< 50\%$ of a tree was living, it was classified as dead. Height of the most-used emergence point of the roost and total height of the tree were measured using a clinometer (PM-10/360PCL, Suunto Company, Helsinki, Finland). Canopy closure was determined using a concave densiometer (Forest Densiometers, Bartlesville, OK); measurements were taken initially at the base of and 5 m away from the roost, in each of the four cardinal directions, for a total of eight readings (Gardner et al., 1991; Lemmon, 1956). However, preliminary comparisons of the 5-m readings and those taken at the base indicated no significant differences ($P > 0.10$); hence, final analyses were performed using only readings taken at the base of the tree. Trees had fully leafed out

before field work began in the last week of May of each year.

Differences in continuous variables were analyzed using the two-tailed Mann-Whitney (Wilcoxon) test with the standard normal deviate Z or the χ^2 approximation to the Kruskal-Wallis test, depending on number of treatments. Cochran's Q test was used for categorical analyses of dichotomous variables when sample sizes were equal, and chi-square tests were used for more complex contingency tables. Correlations were examined with Spearman's test. Level of rejection for all tests was $P \leq 0.05$, except for multiple-comparison (Mann-Whitney) tests performed after a Kruskal-Wallis test; alpha for these tests was adjusted according to Bonferroni procedures. Statistical analyses were performed using SPSS (Norusis, 1990), and data are expressed as $\bar{X} \pm SE$.

RESULTS

Roosting habitat.—Radiotransmitters were placed on 12 northern bats, including one adult female that was not palpably pregnant (captured on 26 May), five palpably pregnant females, three lactating females, one postlactating female, and two juvenile females. Generally only one bat carried a transmitter at any one time. One lactating bat could not be located following placement of its transmitter, but the other transmitters remained attached and functional for an average of 5.6 days (range = 2–9 days). Those 11 radiotagged *M. septentrionalis* eventually were tracked to 32 roost trees.

All roost trees were located in wetlands, and the base of each trunk was submerged in water at some point during both years, either due to seasonal flooding or permanent inundation. Twenty-nine of the 32 roost trees were concentrated within a 20-ha site that was part of property owned by the Borntrager family. The entire roosting area on the Borntrager property had been forested pasture at one time. Although part of the area still was used as pasture in 1993–1994, some of the land was flooded permanently ca. 15 years ago, leading to the death of most trees in the flooded area

(Kurta et al., 1993a, 1996). The roosting area was bounded by riparian woods along the Thornapple River and Lacey Creek to the west, pasture and Vermontville Highway to the north, a cornfield and Bradley Road to the east, and an abandoned railroad track and more pasture to the south. The other roost trees were in two forested wetlands; one roost was 800 m N of the Borntrager property, and the last two were 2 km SW of the Borntrager property. Greatest distance between any two of the 32 roost trees was ca. 2.55 km, but maximum distance between roost trees on the Borntrager property alone was 632 m. Greatest distance from the Thornapple River or Lacey Creek to any roost was 183 m. Two-laned roads, paved or gravel, were <0.7 km from each roost.

Woody stems ≥ 10 cm in diameter were identified to species in plots surrounding 26 roost trees; to preserve independence of the data, separate surveys were not completed on six roost trees because of significant overlap with neighboring roosts. Twenty-three plots were on the Borntrager property; one was in the northern wetland, and two were in the southwestern wetland. A total of 562 trees representing 14 species was identified. The most common species of trees were green ash (40%), silver maple (26%), and American elm (24%); no other species contributed >3% to the total (Foster, 1997).

Characteristics of roost trees.—Roost trees belonged to three species (Table 1). Eighteen roosts (56%) were silver maples, one (3%) was red maple (*A. rubrum*), and 13 (41%) were green ashes. Silver maples and the red maple were combined under "maples" in all analyses, due to structural similarities and the small sample of red maples.

Among the 32 roost trees, mean diameter was 65 ± 1 cm, and mean height was 23.3 ± 0.2 m; diameter and height were correlated ($r_s = 0.77$; $P = 0.0001$). Height of the bats' exit averaged 10.7 ± 0.04 m and was correlated with height of the tree ($r_s = 0.55$;

$P = 0.002$). Height of the exit as a proportion of the height of the tree ranged from 0.12 to 0.98, with a mean of 0.48 ± 0.04 ; that proportion was not correlated with height of the tree ($r_s = 0.04$; $P = 0.82$). There was no significant difference among the four cardinal directions in amount of canopy cover ($\chi^2 = 0.33$, $d.f. = 3$, $P > 0.10$), and the mean was $44 \pm 7\%$. Canopy cover was bimodally distributed, with 13 roosts having 0–20% canopy cover, and nine roosts having 81–100% canopy cover ($\chi^2 = 13.63$, $d.f. = 4$, $P = 0.009$; Fig. 1). There was no correlation between amount of canopy cover and diameter ($r_s = -0.11$, $P = 0.56$), date on which the roost was discovered ($r_s = -0.25$, $P = 0.12$), or maximum population size observed at each roost ($r_s = -0.10$, $P = 0.60$).

There were differences between maples and ashes that were used as roosts. Mean diameter of maples was 83 ± 6 cm compared to 39 ± 0.1 cm for green ashes ($Z = -4.32$, $P < 0.001$). Average height of maples was 26.0 ± 1.2 m but 19.3 ± 1.9 m for green ashes ($Z = -2.74$, $P = 0.006$). Height of the exits, in contrast, did not differ ($Z = -1.05$, $P = 0.29$); maples averaged 11.7 ± 1.1 m compared to 9.6 ± 1.3 m for ashes. Although there was considerable overlap (Table 1), mean canopy cover was greater at roosts in maples ($52 \pm 8\%$) than in green ashes ($40 \pm 11\%$; $Z = -2.07$, $P = 0.04$).

Twenty-nine roosts were observed during emergence, allowing us to determine where bats were roosting (Table 1). Bats roosted underneath exfoliating bark at 15 trees and rested inside hollows or crevices (hereafter combined as "cavities") at the other 14 trees. Roosts in cavities or under bark were split evenly within the maples (eight cavity and eight bark) and within the green ash (six cavity and seven bark). Six of 14 roosts in cavities (four in maples and two in ashes) were crevices or hollows within the main portion of the bole; one roost in a green ash was a crevice formed at the top of a broken trunk, and seven roosts in maples were

TABLE 1.—*Characteristics of roost trees of the northern bat in southern Michigan.*

Roost	Species of tree	Diameter of tree (cm)	Height of tree (m)	Height of exit (m) ^a	Type of roost ^a	Amount of bark ^b	Cavities present	Dead	Canopy cover (%) ^c
A93	Silver maple	49	21.3	11.4	Cavity	Low	Yes	No	84.6
B93	Silver maple	97	32.5			Low	No	No	77.8
C93	Silver maple	76	22.7	11.1	Bark	Medium	Yes	No	89.1
D93	Silver maple	71	24.1	9.6	Cavity	Medium	Yes	No	63.5
E93	Silver maple	110	36.7	17.7	Cavity	High	Yes	Yes	0.0
F93	Green ash	47	26.3	13.8	Cavity	High	Yes	Yes	0.0
G93	Silver maple	64	18.8	5.1	Bark	High	Yes	Yes	0.0
H93	Green ash	44	17.2	12.0	Cavity	Low	Yes	No	81.4
I93	Green ash	39	13.2	12.9	Cavity	Low	Yes	No	85.1
J93 ^d	Silver maple	89	26.5			Low	Yes	No	85.1
K93	Silver maple	81	27.9	11.8	Cavity	Low	Yes	Yes	0.0
L93 ^d	Silver maple	88	25.9	14.9	Cavity	Low	Yes	No	95.3
M93	Green ash	26	18.1	11.7	Cavity	Zero	Yes	No	10.9
N93	Green ash	28	10.0	1.2	Bark	High	No	Yes	0.0
O93	Red maple	54	27.1	17.5	Cavity	Low	Yes	No	74.1
P93	Silver maple	95	31.4	13.3	Bark	High	No	Yes	0.0
Q93	Silver maple	85	26.9	17.7	Bark	Low	Yes	No	64.1
R93	Silver maple	73	21.7	5.7	Bark	High	Yes	Yes	45.9
A94 ^d	Green ash	61	26.1	14.4	Cavity	Low	Yes	No	93.3
B94	Silver maple	105	30.0			Low	Yes	No	71.4
C94	Green ash	48	29.6	15.0	Bark	High	No	Yes	40.9
D94	Green ash	42	20.5	8.5	Bark	Low	No	Yes	0.0
E94	Silver maple	90	23.3	9.0	Bark	Medium	Yes	Yes	59.3
F94	Silver maple	112	29.1	14.3	Bark	Low	Yes	Yes	40.9
G94	Green ash	33	20.0	5.4	Cavity	Low	Yes	Yes	8.0
H94	Silver maple	34	23.1	17.3	Cavity	Low	Yes	No	10.3
I94	Green ash	40	23.0	14.8	Bark	Low	Yes	No	83.4
J94	Green ash	28	9.6	6.8	Bark	Low	Yes	Yes	0.0
K94	Green ash	36	26.1	4.1	Bark	High	No	Yes	4.0
L94	Green ash	28	10.5	3.7	Bark	Medium	Yes	No	19.0
M94	Silver maple	54	15.0	6.3	Bark	Low	Yes	Yes	27.1
N94	Silver maple	150	30.5	4.6	Cavity	Low	Yes	No	86.9

^a Missing data are due to the absence of emergence observations.^b Categories follow Gardner et al. (1991).^c Each value represents the mean of four readings, with one taken in each of the four cardinal directions at the base of each tree.^d Located off the Borntrager property.

crevices at the end of large broken branches. Seventeen trees were living (11 maple and 6 ash), and 15 were dead (eight maple and seven ash). Analysis comparing proportions of roost trees among different species (maple versus ash), types of roost (cavity versus bark), and status (living versus dead) indicated no significant differences ($Q = 0.24$, $d.f. = 2$, $P = 0.89$). There was no difference in amount of loose bark between trees used as cavity roosts and those

used as bark roosts or between maples and ashes ($P > 0.1$; Table 1).

Comparison with a random sample of trees.—We compared roost trees to a random sample of trees that appeared suitable for roosting (Betts, 1997). A suitable tree was defined as one ≥ 10 cm in diameter that contained obvious hollows or crevices or possessed exfoliating bark under which bats might roost. To minimize confounding effects of differences in stand composition,

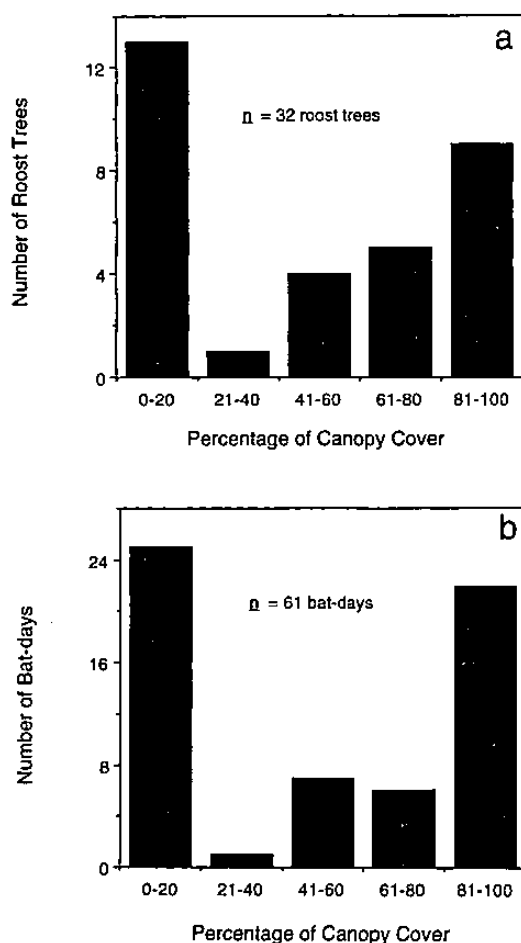


FIG. 1.—Relationship between canopy cover and a) number of roost trees and b) number of bat-days; 1 bat-day represents one radiotracked bat roosting in a tree for 1 day.

we restricted the analysis to the 29 roosts located on the Borntrager property and chose the random sample from trees within the 23 associated circular plots. We used a table of random numbers to choose 46 suitable but unused trees, two from each plot. If two suitable trees were not available in a plot, we randomly chose a replacement from the entire pool of suitable trees. To simplify analysis, only red and silver maples, green ashes, and American elms were chosen because they made up ca. 90% of all trees in the 0.1-ha plots. Although not all parameters were measured for trees

TABLE 2.—Diameter, number of trees alive, and number of trees with cavities for 29 roosts and a random sample of 46 potential roost trees, from the Borntrager property.

Category	n	Diameter (cm)		Number alive	Number with cavities
		$\bar{X}$	SE		
Roosts					
Maples	17	84	7	9	15
Ashes	12	37	2	5	8
Random sample					
Maples	14	62	6	5	11
Ashes	16	36	3	7	11
Elms	16	15	1	0	0

within the plots, we did note species, diameter, amount of bark, presence of cavities, and if the tree was living or dead.

Although suitable elms (i.e., those with cavities or loose bark) made up 35% of the random sample (Table 2), northern bats did not use this species for roosting ($\chi^2 = 13.7$, $d.f. = 2$, $P = 0.001$). Overall, diameter differed significantly among the five potential categories—roost maples, random maples, roost ashes, random ashes, and random elms ($\chi^2 = 55.5$, $d.f. = 4$, $P < 0.0001$; Table 2). Multiple comparisons indicated that elms were smaller than each of the four groups of maple and ash (Mann-Whitney tests, $P < 0.0001$). There was no significant difference in diameter between roost and unused maples ($Z = -2.03$, $P = 0.04$, which was not significant because a Bonferroni correction reduced alpha to 0.008) or between roost and unused ash ($Z = 0.26$, $P = 0.80$). The overall sample of roost trees (all species combined) did not vary from the random sample in terms of the proportion of trees that were alive, the proportion that had cavities, or the proportion that had zero, low, medium, or high amounts of loose bark ($P \geq 0.08$).

Population size.—Mean population size was 17 ± 2 bats ($n = 65$ observations; Fig. 2a); maximum count at any tree averaged 13 ± 3 bats ($n = 29$ roosts). The largest

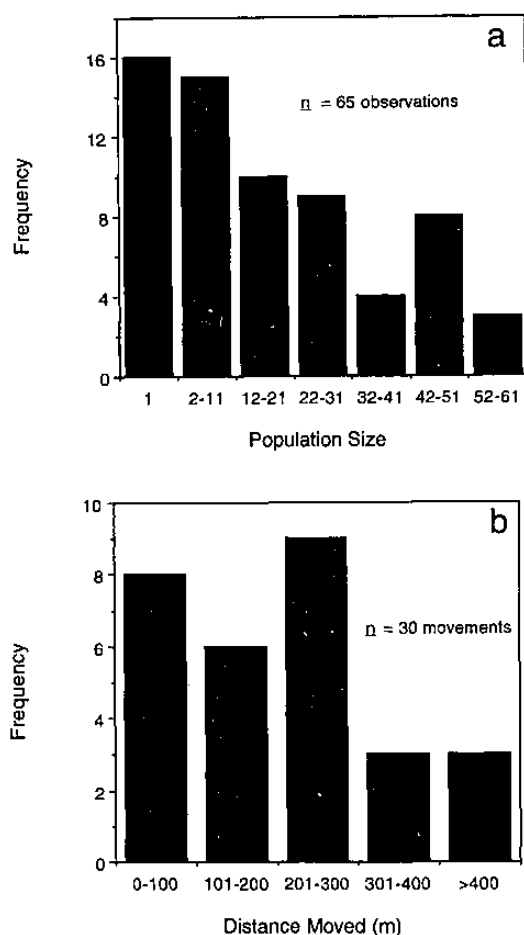


FIG. 2.—a) Population size determined by counting emerging bats at dusk and b) distance moved by each bat that changed roosts. All movements >400 m involved sites off the Borntreger property.

populations were observed at silver maples—A93, G93, and K93—with maximum counts of 60, 43, and 38 bats, respectively. Roost A93 had the 10 highest emergence counts, ranging from 43 to 60 bats; those high counts occurred even though A93 was used only in the last week of May and the first week of June of both years, which was well before flying juveniles entered the volant population and inflated the population size.

Northern bats roosted with other bats more often than roosting alone ($\chi^2 = 18.84$,

$d.f. = 1$, $P < 0.001$; Fig. 2a). Nevertheless, only a single bat left the roost tree during 16 (25%) emergences, and 8 of the 11 bats roosted alone on ≥ 1 day. Two lactating bats and one pregnant individual always roosted with other bats, but they were tracked for only 2–4 days each.

Fidelity to roosts and movements.—One banded bat used the same tree (A93) in 1993 and 1994, but only three roosts (A93, E93, and G93) were used by different radiotagged bats in both years. Northern bats frequently changed trees within each season, averaging 2.8 switches/bat (range = 1–6). Thirty-one changes by the 11 bats occurred over 61 days of observation, indicating that northern bats typically changed roosts every 2 days. Northern bats generally moved to new trees, and only two bats in the 1st year and one bat in the 2nd year ever returned to a tree that the animal had used previously while it was being radiotracked. Mean number of different trees used by each bat was 3.6 (range = 2–7). Number of switches per bat ($r_s = 0.76$, $P = 0.006$) and the number of different trees per bat ($r_s = 0.74$, $P = 0.009$) were correlated with the number of days that the animal was tracked.

Distance that a bat moved was measured for 30 of 31 changes (Fig. 2b; for one change, the animal left to forage before the roost tree was located). Longest distance between two roost trees used by the same bat was ca. 2 km, and shortest distance was 6 m. Average distance moved was 333 ± 88 m, including three switches each ≥ 850 m involving the two sites off the Borntreger property. If only trees on the Borntreger property were included, average movement for the remaining 27 changes was 191 ± 22 m; most (62%) were <250 m. To avoid problems with nonindependence, we also calculated mean distance moved between roosts using average distance traveled by each bat. That mean distance was 414 ± 172 m ($n = 11$ bats) and 190 ± 36 m ($n = 9$ bats) for switches including and not including trees off the Borntreger property, respectively.

TABLE 3.—Comparison of roost trees, habitat, and behavior of northern bats and Indiana bats on the Borntrager property. Continuous variables are presented as $\bar{X} \pm \text{SE}$ followed by n in parentheses.

Character	Northern bats (This study)	Indiana bats (Kurta et al., 1996)	Statistics
Roosting area	ca. 20 ha	ca. 5 ha	
Species of tree	17 maple; 12 ash	0 maple; 23 ash	$\chi^2 = 20.03; P < 0.001$
Type of roost	14 cavity; 15 bark	0 cavity ^a ; 23 bark	$\chi^2 = 15.19; P < 0.001$
Alive or dead	14 alive; 15 dead	0 alive; 23 dead	$\chi^2 = 15.19; P < 0.001$
Canopy cover			
0–50%, 51–100%	17, 12	23 ^b , 0	$\chi^2 = 13.53; P < 0.001$
Diameter of tree (cm)			
All species	63 $\pm$ 6 (29)	41 $\pm$ 1 (23)	$Z = -2.38; P = 0.02$
Maples	82 $\pm$ 7 (17)		
Ashes	37 $\pm$ 2 (12)	41 $\pm$ 1 (23)	$Z = -1.52; P = 0.13$
Total height of tree (m)			
All species	23.0 $\pm$ 0.2 (29)	25.1 $\pm$ 1.6 (23)	$Z = 0.84; P = 0.40$
Maples	26.0 $\pm$ 1.3 (17)		
Ashes	18.7 $\pm$ 2.0 (12)	25.1 $\pm$ 1.6 (23)	$Z = 2.12; P = 0.03$
Height of bat's exit (m)			
All species	10.5 $\pm$ 0.9 (27)	9.9 $\pm$ 0.9 (21)	$Z = -0.45; P = 0.66$
Maples	11.5 $\pm$ 1.2 (15)		
Ashes	9.2 $\pm$ 1.4 (12)	9.9 $\pm$ 0.9 (21)	$Z = -0.34; P = 0.74$
Amount of loose bark ^c			
High, medium, low	6, 3, 6	12, 9, 2	$\chi^2 = 5.56; P > 0.1$
Frequency of changes	1/2 days	1/2.9 days	
Distance moved (m)			
All changes	191 $\pm$ 22 (27)	74.2 $\pm$ 8 (37)	$Z = 4.28; P < 0.001$
Mean value/bat	190 $\pm$ 36 (9)	79.0 $\pm$ 11 (17)	$Z = 2.37; P = 0.02$

^a A few fecal pellets were present in a cavity of one roost tree (Kurta et al., 1993a), but bats were only observed leaving from beneath adjacent bark.

^b All 23 trees were essentially unshaded, with canopy cover likely <20% at each.

^c Only trees at which bats actually roosted under bark; categories of low to high follow Gardner et al. (1991).

Comparison with the Indiana bat.—Kurta et al. (1996) discovered 23 roosts used by adult female and juvenile Indiana bats over 3 summers (1991–1993) compared with the 32 roosts that we located for the northern bat during 1993–1994. Two trees used by female Indiana bats were each used on 1 day by northern bats; however, the two species of bat never were found in the same tree on the same day (not even within the same year at one tree), and the exact roosting site on the tree also differed (Foster, 1997). All 23 roosts of the Indiana bat were on the Borntrager property, as were 29 roosts of the northern bat (Table 3). We restricted comparisons to roosts on the Born-

trager property to minimize stand differences; nevertheless, inclusion of the three roosts located off that property did not change the significance of any test.

Roosts of the Indiana bat were tightly grouped into a 5-ha flooded area that was totally within the more heterogeneous 20-ha site that contained roosts of the northern bat. Although height and diameter of maples and ashes differed for the northern bat, ashes used by northern bats were similar in diameter to, but slightly shorter than, those used by Indiana bats (Table 3). Heights of the exits used by the two species did not differ. The amount of loose bark on trees used as bark roosts did not differ between

species of bats (Table 3) or from samples of available trees for either species of bat (Kurta et al., 1996; this study).

Despite similarities, there were many differences in roosting behavior (Table 3). Adult female and juvenile Indiana bats were radiotracked only to green ashes, but northern bats were found most often in maples. All trees used by Indiana bats were dead (as defined in the present study), but those used by northern bats were equally divided among living and dead trees. Indiana bats always emerged from underneath loose bark, but northern bats roosted in cavities or under bark in about equal proportions. Canopy cover at roosts of the Indiana bat was invariably <20%, but 17 of 29 roosts of the northern bat were more heavily shaded. Average distance moved by northern bats when changing trees was more than double that of the Indiana bat.

DISCUSSION

Population size.—In summer, the northern bat generally is colonial, but reproductive females and juveniles often roost alone (Fig. 2a)—a pattern similar to that seen in the Indiana bat (Kurta et al., 1996). On 27 May 1993, we counted 60 adult northern bats leaving the hollow of roost A93, and nine other observations of this tree yielded emergence counts of >42 bats, making this the largest nonhibernating group ever reported for the species. Sasse and Pekins (1996) watched 36 northern bats leave a roost on an unspecified date in New Hampshire, and Cope and Humphrey (1972) noted 34 adults at a barn in Indiana on 22 June 1966.

In both years, the largest emergence counts occurred early in the season at roost A93, a living maple with 85% canopy cover; population size then rapidly fell to <20 animals during most counts after mid-June. Sasse and Pekins (1996) also reported higher population counts of northern bats early in the maternity season, before parturition. A large number of bats per roost early in the year may be related to thermoregulatory

needs, because clustering by bats and other small homeotherms should allow them to maintain a high body temperature with less energetic expenditure, especially with the added insulation of a wooden cavity (du Plessis and Williams, 1994; Kurta, 1985, 1986). However, if energetic constraints were the main reason for these large groupings early in the maternity season, one might not expect bats to roost in a tree with heavy canopy cover (A93; Table 1), which would reduce solar warming during the coolest part of the maternity season. Another possibility for large colonies early in the year, when air temperatures are low and prey are scarce, is that such large groupings may facilitate the transfer of information among bats concerning the location of productive foraging areas and thereby increase foraging efficiency of individuals (Sparling and Krapu, 1994; Wilkinson, 1992).

The number of northern bats presumably swelled to >100 individuals in this "colony" in July, after young became volant, because litter size typically is one (Cope and Humphrey, 1972), and preweaning mortality in colonial insectivorous bats generally is <12% (Tuttle and Stevenson, 1982). Obviously all members of this colony did not roost together during the remainder of the summer, because maximum population size after 4 July of either year, when volant juveniles first were captured, was only 38 bats and most counts were <20 bats (Foster, 1997). This suggested that some members of the group dispersed to different areas, or if bats remained nearby, they used a large number of different trees at any one time.

Fidelity to roosts and movements.—Northern bats changed roosts about every 2 days, which appears typical of tree-roosting bats (Lewis, 1995). For example, our data on roost switching by northern bats were similar to those reported for Indiana bats (2.9 days—Kurta et al., 1996), silver-haired bats, *Lasionycteris noctivagans* (ca. 1 day—Mattson et al., 1996), and long-legged bats, *M. volans* (2.2 days—Ormsbee, 1996). Northern bats generally switched to

a new tree to which they had not been tracked previously. They used 3.6 different trees on average, and some bats used as many as seven different trees; however, the number of trees used was correlated with the number of days that the transmitter remained functional. This implied that each bat would have used a much larger, but unknown, number of trees over the course of an entire maternity season, as reported for Indiana bats (Kurta et al., 1996) and long-legged bats (Ormsbee, 1996). The need for a large number of trees by these bats is extremely important from a management perspective, especially in the agricultural landscapes of midwestern United States, where forested areas often are young, widely dispersed, and limited in extent.

Characteristics of the roosts and habitat.—There has been only one other study of the northern bat that involved radiotracking. Sasse and Pekins (1996) surveyed roosting habits of the northern bat in the White Mountain National Forest in northern New Hampshire during the same 2 years as our study (1993–1994). Unfortunately, it was difficult to compare our results to those of Sasse and Pekins (1996) because of differing methodology and because they apparently restricted their statistical analysis to a subset of their roost trees (snags). Nevertheless, our study and that of Sasse and Pekins (1996) showed that the northern bat was primarily a tree-dwelling species, despite earlier reports of this bat using man-made structures (Brandon, 1961; Cope and Humphrey, 1972).

Sasse and Pekins (1996:97) “presumed” that the northern bat preferred totally dead trees (snags), as do most species that use cavities or bark (Barclay and Brigham, 1996; Brigham et al., 1997; Campbell et al., 1996; Kurta et al., 1993a, 1996). However, 13 of 47 roosts in New Hampshire (Sasse and Pekins, 1996) and 17 of 32 roosts in Michigan (Table 1) were in living trees. This suggests that choice of a living tree is common among northern bats compared

with other species of North American bats with similar roosting habits.

Sasse and Pekins (1996) did not determine if their bats were roosting underneath bark or in cavities, but northern bats in Michigan were generalized in their roosting behavior, with about one-half of the roosts in cavities and the other one-half under exfoliating bark (Table 1). Breadth of the roosting niche of a population is determined by variation among and within individuals (i.e., a generalist population may result because different members of the population specialize in different ways or because some or all individuals are generalists—C. Gubbins, in litt.; Heinrich, 1976). In our study, five of 11 bats that were radiotracked roosted in cavities and under bark, suggesting that this generalized roosting pattern is a result of variation within individuals.

Northern bats in southern Michigan roosted mostly in silver maple (56%) and green ash (41%), with a single roost in red maple (3%). Our study is the first to report the use of silver maple and green ash by this bat. Mumford and Cope (1964) found a colony of northern bats living under the bark of an American elm in Indiana, and G. Gardner (in litt.) discovered one under the exfoliating bark of a slippery elm (*U. rubra*) in Illinois. Clark et al. (1987) reported northern bats under the bark of an unspecified elm in Iowa. Sasse and Pekins (1996), in contrast, found most of their 47 roosts in American beech (30%), sugar maple (28%), yellow birch (*Betula alleghaniensis*—17%), and red maple (13%). Most likely, northern bats are not dependent on a particular species of tree throughout their range. Rather, certain species of trees tend to form suitable cavities or retain bark and will be used whenever they are present. Different ecological equivalents likely occur in different habitats and geographic areas, and northern bats will use whatever appropriate species of trees that happens to be there (Gardner et al., 1991; Mattson et al., 1996).

Despite roosting in elms elsewhere in the

Midwest, northern bats in Michigan did not use them, even though 34.7% of potential roost trees (i.e., those with cavities or peeling bark) were elms (Table 2). Tree-roosting bats often select trees that are larger in diameter than the mean for all available trees (Kurta et al., 1996; Sasse and Pekins, 1996; Taylor and Savva, 1988; Vonnhof and Barclay, 1996), possibly because large trees provide more insulation or a larger space in which bats may cluster, thereby reducing thermoregulatory costs (Gardner et al., 1991; Kurta, 1985, 1986). However, Dutch elm disease is common in southern Michigan, and many elms succumb before reaching even moderate size (Barnes and Wagner, 1981). Presumably the lack of roosts in elms was related to their extremely small size and the absence of cavities in such young trees (Table 2) and not to specific structural characteristics of elms in Michigan. Other than failing to use elms in proportion to their abundance, northern bats did not appear to select their roost tree on the basis of any of the observed parameters.

After comparing roosts in maples and ashes, we found that the two groups did not differ in amount of loose bark, proportion that was alive or dead, or proportion that had cavities. Mean height of the tree was significantly greater in maples than ashes, but this had no noticeable effect on the bats, because the height at which bats roosted (as indicated by height of the exit) did not differ between maples and ashes. Although average diameter of maples was more than double that of ashes, this also may not be biologically significant because of the different growth form of the two types of trees. Ashes generally have a single trunk, whereas maples frequently divide into multiple trunks ca. 2 m above the ground (i.e., above the height at which diameter is typically measured—Barnes and Wagner, 1981). Reports of average diameters of roost trees that lump multiple species of trees with different patterns of growth (Callahan et al., 1997; Sasse and Pekins, 1996) are of questionable use in predicting if a

particular tree will or will not be used by a bat.

Northern bats in Michigan did not require trees low in canopy cover for roosting and exhibited a bimodal choice in percentage of canopy cover (Fig. 1). Most bat-days were spent in trees with either 0–20% or 81–100% canopy cover, and this likely reflects the fact that about one-half of the roost trees were living and one-half were considered dead (Table 1). Northern bats were not forced to use trees with high shade because permanently flooded areas of the site contained hundreds of apparently suitable unshaded trees (Kurta et al., 1996). That low canopy cover is not required by northern bats also is evident in the data of Sasse and Pekins (1996), who reported an average of 83% canopy cover for their snags—a value similar to the forest surrounding their roosts.

Vonnhof and Barclay (1996) showed that some bats in western Canada favored roosts with low canopy cover. They speculated that such a clear space would provide easier access for bats (less chance of accidents), make the roost more visible in the landscape, and decrease predation, because bats could fly directly to and from the roost without maneuvering. The bimodal nature of our data (Fig. 1) is not consistent with the various hypotheses proposed by Vonnhof and Barclay (1996). It may be that, for northern bats, an average value for clutter around a tree is less important than clutter directly in front of the opening to the roost (Betts, 1997). In addition, the northern bat is a gleaning species (Faure et al., 1993) and should be able to maneuver around clutter better than similar-sized bats (Norberg and Rayner, 1987).

Interspecific comparisons.—We did not analyze as many characters as other comparative studies in North America (primarily between silver-haired and big brown bats in heavily forested areas of the West—Betts, 1996; Vonnhof, 1996; Vonnhof and Barclay, 1996), but we detected differences in the roosting behavior of Indiana and

northern bats. In southern Michigan, the northern bat had a distinctly broader roosting niche, utilizing maples, cavities, living trees, and sites with higher canopy cover more often than its congener, the Indiana bat (Table 3). Those differences were detected at a site where the number of trees that appeared suitable for roosting was very high (97/ha in the flooded area alone—Kurta et al., 1996), and trees that were used presumably reflected preferences of each species at this latitude. The fact that both species remained in such a small area over multiple years also suggested that roosting conditions were quite favorable and, perhaps, optimum.

Although the roosting niche of these two species differed, they showed considerable overlap at our study site (Table 3), and the fact that Indiana bats occasionally have used cavities in other areas (Callahan, 1997; Kurta et al., 1993b) indicates further similarity in what the two species find acceptable. The overlap in roosting niches suggests the possibility of interspecific interactions in areas where suitable roosts are not as abundant as on the Borntreger property, and this could be especially important for the management of the endangered Indiana bat, because its breeding range is centered on the agricultural Midwest—a region where, on average, suitable trees are likely low in number and widely scattered. To gain further insight into the roosting requirements of Indiana and northern bats, as well as potential interactions between them, we suggest that future studies simultaneously examine the roosting niche of each species in areas where they do not coexist, where the composition of the forest differs, and where the abundance of suitable trees differs.

In addition, previous comparative reports from North America have concentrated on gross characteristics of roost trees and their surrounding habitat (Betts, 1996; Vonhof, 1996; Vonhof and Barclay, 1996; this study). Future projects also should compare aspects of the roosting site itself (e.g., di-

mensions of cavities and openings, diameter of the trunk and canopy cover at the actual roosting site, thickness and insulative value of wood or bark, etc.). Quantifying such microcharacters (Robb et al., 1996; Tidemann and Flavel, 1987) will be difficult but ultimately may provide better insight into requirements of coexisting species and how their roosting niche is differentiated.

ACKNOWLEDGMENTS

We thank the Borntreger for access to their property. J. Caryl, J. Cypher, K. Parish-Foster, and B. Wachoawiak provided assistance in the field. D. Hart provided temporary financial assistance to R. W. Foster to begin the project. Additional funding was provided by a Meta Hellwig Graduate Fellowship to R. W. Foster and by grants from the Nongame Wildlife Fund of the Michigan Department of Natural Resources, National Science Foundation (USE-8952309), and Eastern Michigan University Graduate School to A. Kurta. Data from this paper contributed to a thesis submitted by R. W. Foster to the Graduate School of Eastern Michigan University in partial fulfillment of requirements for the M.S. degree. C. Bach, J. Eisenbach, S. W. Murray, and J. O. Whitaker, Jr. provided helpful comments.

LITERATURE CITED

- ALBERT, D. A., S. R. DENTON, and B. V. BARNES. 1986. Regional landscape ecosystems of Michigan. School of Natural Resources, University of Michigan, Ann Arbor.
- BAKER, R. H. 1983. Michigan mammals. Michigan State University Press, East Lansing.
- BARBOUR, R. W., and W. H. DAVIS. 1969. Bats of America. University Press of Kentucky, Lexington.
- BARCLAY, R. M. R., and R. M. BRIGHAM. (EDS.). 1996. Bats and forests. Research Branch, Ministry of Forests, Victoria, British Columbia, Canada.
- BARNES, B. V., and W. H. WAGNER, JR. 1981. Michigan trees. University of Michigan Press, Ann Arbor.
- BETTS, B. J. 1996. Roosting behaviour of silver-haired bats (*Lasionycteris noctivagans*) and big brown bats (*Eptesicus fuscus*) in northeast Oregon. Pp. 55–61, in Bats and forests (R. M. R. Barclay and R. M. Brigham, eds.). Research Branch, Ministry of Forests, Victoria, British Columbia, Canada.
- . 1997. Roosts used by maternity colonies of silver-haired bats in northeastern Oregon. *Journal of Mammalogy*, 79:643–650.
- BONACCORSO, F. J., and N. SMYTHE. 1972. Punch-marking bats: an alternative to banding. *Journal of Mammalogy*, 53:389–390.

- BRANDON, R. A. 1961. Observations of young keen bats. *Journal of Mammalogy*, 42:400–401.
- BREWER, R., G. A. MCPEEK, AND R. J. ADAMS, JR. 1991. The atlas of breeding birds of Michigan. Michigan State University Press, East Lansing.
- BRIGHAM, R. M., M. J. VONHOF, R. M. BARCLAY, AND J. C. GWILLIAM. 1997. Roosting behavior and roost-site preferences of forest-dwelling California bats (*Myotis californicus*). *Journal of Mammalogy*, 78:1231–1239.
- CALLAHAN, E. V., R. D. DROBNEY, AND R. L. CLAWSON. 1997. Selection of summer roosting sites by Indiana bats (*Myotis sodalis*) in Missouri. *Journal of Mammalogy*, 78:818–825.
- CAMPBELL, L. A., J. G. HALLETT, AND M. A. O'CONNELL. 1996. Conservation of bats in managed forests: use of roosts by *Lasionycteris noctivagans*. *Journal of Mammalogy*, 77:976–984.
- CLARK, B. S., J. B. BOWLES, AND B. K. CLARK. 1987. Summer occurrence of the Indiana bat, Keen's myotis, evening bat, silver-haired bat and eastern pipistrelle in Iowa. *Proceedings of the Iowa Academy of Science*, 94:89–93.
- COPE, J. B., AND S. R. HUMPHREY. 1972. Reproduction of the bats *Myotis keenii* and *Pipistrellus subflavus* in Indiana. *Bat Research News*, 13:9.
- DU PLESSIS, M. A., AND J. B. WILLIAMS. 1994. Communal cavity-roosting in cooperatively-breeding green woodhoopoes: consequences for energy expenditure and the seasonal pattern of mortality. *The Auk*, 111:292–299.
- FAURE, P. A., J. H. FULLARD, AND J. W. DAWSON. 1993. The gleanings attacks of the northern long-eared bat, *Myotis septentrionalis*, are relatively inaudible to moths. *Journal of Experimental Biology*, 178:173–189.
- FOSTER, R. W. 1997. Roost-site selection by the northern bat (*Myotis septentrionalis*) in Michigan. M.S. thesis. Eastern Michigan University, Ypsilanti.
- GARDNER, J. E., J. D. GARNER, AND J. E. HOFMANN. 1991. Summer roost selection and roosting behavior of *Myotis sodalis* (Indiana bat) in Illinois. Illinois Natural History Survey, Bloomington.
- GERRELL-LUNDBERG, K., AND R. GERRELL. 1994. The mating behaviour of the pipistrelle and the Nathusius' pipistrelle (Chiroptera)—a comparison. *Folia Zoologica*, 43:315–324.
- HARMATA, W. 1969. Kolonia letnia nietoperza *Myotis bechsteini* (Kuhl) w Szymbarku koło Gorlic w woj. rzeszowskim. Uwagi o biologii i występowaniu. *Przegląd Zoologiczny*, 13:233–238.
- HEINRICH, B. 1976. Foraging specializations of individual bumblebees. *Ecological Monographs*, 46:105–128.
- HUMPHREY, S. R. 1975. Nursery roosts and community diversity of Nearctic bats. *Journal of Mammalogy*, 56:321–346.
- KALCOUNIS, M. C., AND K. R. HECKER. 1996. Intraspecific variation in roost-site selection by little brown bats (*Myotis lucifugus*). Pp. 81–90, in *Bats and forests* (R. M. R. Barclay and R. M. Brigham, eds.). Research Branch, Ministry of Forests, Victoria, British Columbia, Canada.
- KURTA, A. 1985. External insulation available to a non-nesting mammal, the little brown bat (*Myotis lucifugus*). *Comparative Biochemistry and Physiology, A. Comparative Physiology*, 82:413–420.
- . 1986. Factors affecting the resting and post-flight body temperature of little brown bats, *Myotis lucifugus*. *Physiological Zoology*, 59:429–438.
- KURTA, A., AND R. FOSTER. 1995. The brown creeper (Aves: Certhiidae): a potential competitor of bark-roosting bats? *Bat Research News*, 36:6–7.
- KURTA, A., K. J. WILLIAMS, AND R. MIES. 1996. Ecological, behavioural, and thermal observations of a peripheral population of Indiana bats (*Myotis sodalis*). Pp. 102–117, in *Bats and forests* (R. M. R. Barclay and R. M. Brigham, eds.). Research Branch, Ministry of Forests, Victoria, British Columbia, Canada.
- KURTA, A., D. KING, J. A. TERAMINO, J. M. STRIBLEY, AND K. J. WILLIAMS. 1993a. Summer roosts of the endangered Indiana bat (*Myotis sodalis*) on the northern edge of its range. *The American Midland Naturalist*, 129:132–138.
- KURTA, A., J. KATH, E. L. SMITH, R. FOSTER, M. W. ORICK, AND R. ROSS. 1993b. A maternity roost of the endangered Indiana bat (*Myotis sodalis*) in an unshaded, hollow, sycamore tree (*Platanus occidentalis*). *The American Midland Naturalist*, 130:405–407.
- LEMMON, P. E. 1956. A spherical densiometer for estimating forest overstory density. *Forest Science*, 2:314–320.
- LEWIS, S. E. 1995. Roost fidelity of bats: a review. *Journal of Mammalogy*, 76:481–496.
- LUNNEY, D., J. BARKER, D. PRIDDEL, AND M. O'CONNELL. 1988. Roost selection by Gould's long-eared bat, *Nyctophilus gouldi* Tomes (Chiroptera: Vespertilionidae), in logged forest on the south coast of New South Wales. *Australian Wildlife Research*, 15:375–384.
- MAEDA, K. 1974. Eco-éthologie de la grande noctule, *Nyctalus lasiopterus*, à Sapporo, Japon. *Mammalia*, 38:461–487.
- MASON, C. F., R. E. STEBBINGS, AND G. P. WINN. 1972. Noctules and starlings competing for roosting holes. *Journal of Zoology (London)*, 166:467.
- MATTSON, T. A., S. W. BUSKIRK, AND N. L. STANTON. 1996. Roost sites of the silver-haired bat (*Lasionycteris noctivagans*) in the Black Hills, South Dakota. *The Great Basin Naturalist*, 56:247–253.
- MUMFORD, R. E., AND J. B. COPE. 1964. Distribution and status of the Chiroptera of Indiana. *The American Midland Naturalist*, 72:473–489.
- NORBERG, U. M., AND J. M. V. RAYNER. 1987. Ecological morphology and flight in bats (Mammalia: Chiroptera): wing adaptations, flight performance, foraging strategy, and echolocation. *Philosophical Transactions of the Royal Society of London*, B316:335–427.
- NORRIS, M. J. 1990. SPSS base system user's guide. SPSS, Inc., Chicago, Illinois.
- ORMSBEE, P. C. 1996. Characteristics, use, and distribution of day roosts selected by female *Myotis volans* (long-legged myotis) in forested habitat of the central Oregon Cascades. Pp. 124–131, in *Bats and forests* (R. M. R. Barclay and R. M. Brigham, eds.). Research Branch, Ministry of Forests, Victoria, British Columbia, Canada.

northern bats. In southern Michigan, the northern bat had a distinctly broader roosting niche, utilizing maples, cavities, living trees, and sites with higher canopy cover more often than its congener, the Indiana bat (Table 3). Those differences were detected at a site where the number of trees that appeared suitable for roosting was very high (97/ha in the flooded area alone—Kurta et al., 1996), and trees that were used presumably reflected preferences of each species at this latitude. The fact that both species remained in such a small area over multiple years also suggested that roosting conditions were quite favorable and, perhaps, optimum.

Although the roosting niche of these two species differed, they showed considerable overlap at our study site (Table 3), and the fact that Indiana bats occasionally have used cavities in other areas (Callahan, 1997; Kurta et al., 1993b) indicates further similarity in what the two species find acceptable. The overlap in roosting niches suggests the possibility of interspecific interactions in areas where suitable roosts are not as abundant as on the Borntrager property, and this could be especially important for the management of the endangered Indiana bat, because its breeding range is centered on the agricultural Midwest—a region where, on average, suitable trees are likely low in number and widely scattered. To gain further insight into the roosting requirements of Indiana and northern bats, as well as potential interactions between them, we suggest that future studies simultaneously examine the roosting niche of each species in areas where they do not coexist, where the composition of the forest differs, and where the abundance of suitable trees differs.

In addition, previous comparative reports from North America have concentrated on gross characteristics of roost trees and their surrounding habitat (Betts, 1996; Vonhof, 1996; Vonhof and Barclay, 1996; this study). Future projects also should compare aspects of the roosting site itself (e.g., di-

mensions of cavities and openings, diameter of the trunk and canopy cover at the actual roosting site, thickness and insulative value of wood or bark, etc.). Quantifying such microcharacters (Robb et al., 1996; Tidemann and Flavel, 1987) will be difficult but ultimately may provide better insight into requirements of coexisting species and how their roosting niche is differentiated.

ACKNOWLEDGMENTS

We thank the Borntragers for access to their property. J. Caryl, J. Cypher, K. Parish-Foster, and B. Wachoawiak provided assistance in the field. D. Hart provided temporary financial assistance to R. W. Foster to begin the project. Additional funding was provided by a Meta Hellwig Graduate Fellowship to R. W. Foster and by grants from the Nongame Wildlife Fund of the Michigan Department of Natural Resources, National Science Foundation (USE-8952309), and Eastern Michigan University Graduate School to A. Kurta. Data from this paper contributed to a thesis submitted by R. W. Foster to the Graduate School of Eastern Michigan University in partial fulfillment of requirements for the M.S. degree. C. Bach, J. Eisenbach, S. W. Murray, and J. O. Whitaker, Jr. provided helpful comments.

LITERATURE CITED

- ALBERT, D. A., S. R. DENTON, and B. V. BARNES. 1986. Regional landscape ecosystems of Michigan. School of Natural Resources, University of Michigan, Ann Arbor.
- BAKER, R. H. 1983. Michigan mammals. Michigan State University Press, East Lansing.
- BARBOUR, R. W., and W. H. DAVIS. 1969. Bats of America. University Press of Kentucky, Lexington.
- BARCLAY, R. M. R., and R. M. BRIGHAM. (EDS.). 1996. Bats and forests. Research Branch, Ministry of Forests, Victoria, British Columbia, Canada.
- BARNES, B. V., and W. H. WAGNER, JR. 1981. Michigan trees. University of Michigan Press, Ann Arbor.
- BETTS, B. J. 1996. Roosting behaviour of silver-haired bats (*Lasionycteris noctivagans*) and big brown bats (*Eptesicus fuscus*) in northeast Oregon. Pp. 55–61, in Bats and forests (R. M. R. Barclay and R. M. Brigham, eds.). Research Branch, Ministry of Forests, Victoria, British Columbia, Canada.
- . 1997. Roosts used by maternity colonies of silver-haired bats in northeastern Oregon. *Journal of Mammalogy*, 79:643–650.
- BONACCORSO, F. J., and N. SMYTHE. 1972. Punch-marking bats: an alternative to banding. *Journal of Mammalogy*, 53:389–390.

- BRANDON, R. A. 1961. Observations of young keen bats. *Journal of Mammalogy*, 42:400–401.
- BREWER, R., G. A. McPECK, AND R. J. ADAMS, JR. 1991. The atlas of breeding birds of Michigan. Michigan State University Press, East Lansing.
- BRIGHAM, R. M., M. J. VONHOF, R. M. BARCLAY, AND J. C. GWILLIAM. 1997. Roosting behavior and roost-site preferences of forest-dwelling California bats (*Myotis californicus*). *Journal of Mammalogy*, 78:1231–1239.
- CALLAHAN, E. V., R. D. DROBNEY, AND R. L. CLAWSON. 1997. Selection of summer roosting sites by Indiana bats (*Myotis sodalis*) in Missouri. *Journal of Mammalogy*, 78:818–825.
- CAMPBELL, L. A., J. G. HALLETT, AND M. A. O'CONNELL. 1996. Conservation of bats in managed forests: use of roosts by *Lasionycteris noctivagans*. *Journal of Mammalogy*, 77:976–984.
- CLARK, B. S., J. B. BOWLES, AND B. K. CLARK. 1987. Summer occurrence of the Indiana bat, Keen's myotis, evening bat, silver-haired bat and eastern pipistrelle in Iowa. *Proceedings of the Iowa Academy of Science*, 94:89–93.
- COPE, J. B., AND S. R. HUMPHREY. 1972. Reproduction of the bats *Myotis keenii* and *Pipistrellus subflavus* in Indiana. *Bat Research News*, 13:9.
- DU PLESSIS, M. A., AND J. B. WILLIAMS. 1994. Communal cavity-roosting in cooperatively-breeding green woodhoopoes: consequences for energy expenditure and the seasonal pattern of mortality. *The Auk*, 111:292–299.
- FAURE, P. A., J. H. FULLARD, AND J. W. DAWSON. 1993. The gleaner attacks of the northern long-eared bat, *Myotis septentrionalis*, are relatively inaudible to moths. *Journal of Experimental Biology*, 178:173–189.
- FOSTER, R. W. 1997. Roost-site selection by the northern bat (*Myotis septentrionalis*) in Michigan. M.S. thesis. Eastern Michigan University, Ypsilanti.
- GARDNER, J. E., J. D. GARNER, AND J. E. HOFMANN. 1991. Summer roost selection and roosting behavior of *Myotis sodalis* (Indiana bat) in Illinois. Illinois Natural History Survey, Bloomington.
- GERRELL-LUNDBERG, K., AND R. GERRELL. 1994. The mating behaviour of the pipistrelle and the Nathusius' pipistrelle (Chiroptera)—a comparison. *Folia Zoologica*, 43:315–324.
- HARMATA, W. 1969. Kolonia letnia nietoperza *Myotis bechsteini* (Kuhl) w Szymbarku koło Gorlic w woj. rzeszowskim. Uwagi o biologii i występowaniu. *Przegląd Zoologiczny*, 13:233–238.
- HEINRICH, B. 1976. Foraging specializations of individual bumblebees. *Ecological Monographs*, 46:105–128.
- HUMPHREY, S. R. 1975. Nursery roosts and community diversity of Nearctic bats. *Journal of Mammalogy*, 56:321–346.
- KALCOUNIS, M. C., AND K. R. HECKER. 1996. Intraspecific variation in roost-site selection by little brown bats (*Myotis lucifugus*). Pp. 81–90, in *Bats and forests* (R. M. R. Barclay and R. M. Brigham, eds.). Research Branch, Ministry of Forests, Victoria, British Columbia, Canada.
- KURTA, A. 1985. External insulation available to a non-nesting mammal, the little brown bat (*Myotis lucifugus*). *Comparative Biochemistry and Physiology, A. Comparative Physiology*, 82:413–420.
- . 1986. Factors affecting the resting and post-flight body temperature of little brown bats, *Myotis lucifugus*. *Physiological Zoology*, 59:429–438.
- KURTA, A., AND R. FOSTER. 1995. The brown creeper (Aves: Certhiidae): a potential competitor of bark-roosting bats? *Bat Research News*, 36:6–7.
- KURTA, A., K. J. WILLIAMS, AND R. MIES. 1996. Ecological, behavioural, and thermal observations of a peripheral population of Indiana bats (*Myotis sodalis*). Pp. 102–117, in *Bats and forests* (R. M. R. Barclay and R. M. Brigham, eds.). Research Branch, Ministry of Forests, Victoria, British Columbia, Canada.
- KURTA, A., D. KING, J. A. TERAMINO, J. M. STRIBLEY, AND K. J. WILLIAMS. 1993a. Summer roosts of the endangered Indiana bat (*Myotis sodalis*) on the northern edge of its range. *The American Midland Naturalist*, 129:132–138.
- KURTA, A., J. KATH, E. L. SMITH, R. FOSTER, M. W. ORICK, AND R. ROSS. 1993b. A maternity roost of the endangered Indiana bat (*Myotis sodalis*) in an unshaded, hollow, sycamore tree (*Platanus occidentalis*). *The American Midland Naturalist*, 130:405–407.
- LEMMON, P. E. 1956. A spherical densiometer for estimating forest overstory density. *Forest Science*, 2:314–320.
- LEWIS, S. E. 1995. Roost fidelity of bats: a review. *Journal of Mammalogy*, 76:481–496.
- LUNNEY, D., J. BARKER, D. PRIDDEL, AND M. O'CONNELL. 1988. Roost selection by Gould's long-eared bat, *Nyctophilus gouldi* Tomes (Chiroptera: Vespertilionidae), in logged forest on the south coast of New South Wales. *Australian Wildlife Research*, 15:375–384.
- MAEDA, K. 1974. Eco-éthologie de la grande noctule, *Nyctalus lasiopterus*, à Sapporo, Japon. *Mammalia*, 38:461–487.
- MASON, C. F., R. E. STEBBINGS, AND G. P. WINN. 1972. Noctules and starlings competing for roosting holes. *Journal of Zoology (London)*, 166:467.
- MATTSON, T. A., S. W. BUSKIRK, AND N. L. STANTON. 1996. Roost sites of the silver-haired bat (*Lasionycteris noctivagans*) in the Black Hills, South Dakota. *The Great Basin Naturalist*, 56:247–253.
- MUMFORD, R. E., AND J. B. COPE. 1964. Distribution and status of the Chiroptera of Indiana. *The American Midland Naturalist*, 72:473–489.
- NORBERG, U. M., AND J. M. V. RAYNER. 1987. Ecological morphology and flight in bats (Mammalia: Chiroptera): wing adaptations, flight performance, foraging strategy, and echolocation. *Philosophical Transactions of the Royal Society of London*, B316:335–427.
- NORRIS, M. J. 1990. SPSS base system user's guide. SPSS, Inc., Chicago, Illinois.
- ORMSBEE, P. C. 1996. Characteristics, use, and distribution of day roosts selected by female *Myotis volans* (long-legged myotis) in forested habitat of the central Oregon Cascades. Pp. 124–131, in *Bats and forests* (R. M. R. Barclay and R. M. Brigham, eds.). Research Branch, Ministry of Forests, Victoria, British Columbia, Canada.

- ROBB, J. R., M. S. CRAMER, A. R. PARKER, AND R. P. URBANEK. 1996. Use of tree cavities by fox squirrels and raccoons in Indiana. *Journal of Mammalogy*, 77: 1017–1027.
- SASSE, D. B., AND P. J. PEKINS. 1996. Summer roosting ecology of northern long-eared bats (*Myotis septentrionalis*) in the White Mountain National Forest. Pp. 91–101, in *Bats and forests* (R. M. R. Barclay and R. M. Brigham, eds.). Research Branch, Ministry of Forests, Victoria, British Columbia, Canada.
- SPARLING, D. W., AND G. L. KRAPU. 1994. Communal roosting and foraging behavior of staging sandhill cranes. *The Wilson Bulletin*, 106:62–77.
- TAYLOR, R. J., AND N. M. SAVVA. 1988. Use of roost sites by four species of bats in state forest in south-eastern Tasmania. *Australian Wildlife Research*, 15: 637–645.
- TIDEMANN, C. R., AND S. C. FLAVEL. 1987. Factors affecting choice of diurnal roost site by tree-hole bats (Microchiroptera) in south-eastern Australia. *Australian Wildlife Research*, 14:459–473.
- TUTTLE, M. D., AND D. L. HENSLEY. 1993. The bat house builder's handbook. Bat Conservation International, Austin, Texas.
- TUTTLE, M. D., AND D. STEVENSON. 1982. Growth and survival of bats. Pp. 105–150, in *Ecology of bats* (T. H. Kunz, ed.). Plenum Press, New York.
- VAN HEERDT, P. F., AND J. W. SLUTTER. 1965. Notes on the distribution and behaviour of the noctule bat (*Nyctalus noctula*) in the Netherlands. *Mammalia*, 29:463–477.
- VONHOF, M. J. 1996. Roost-site preferences of big brown bats (*Eptesicus fuscus*) and silver-haired bats (*Lasionycteris noctivagans*) in the Pend d'Oreille Valley in southern British Columbia. Pp. 62–80, in *Bats and forests* (R. M. R. Barclay and R. M. Brigham, eds.). Research Branch, Ministry of Forests, Victoria, British Columbia, Canada.
- VONHOF, M. J., AND R. M. R. BARCLAY. 1996. Roost-site selection and roosting ecology of forest-dwelling bats in southern British Columbia. *Canadian Journal of Zoology*, 74:1797–1805.
- WILKINSON, G. S. 1992. Information transfer at evening bat colonies. *Animal Behaviour*, 44:501–518.

Submitted 25 March 1998. Accepted 23 June 1998.

Associate Editor was C. Renn Tumlison.